15. தீர்வு:

$$k = \frac{2.303}{t} \log \frac{[A_0]}{[A]}$$

$$k = \frac{2.303}{1\,min} \log \frac{[0.08]}{[0.04]}$$

$$k = 2.303 \log 2$$

$$k = 2.303 \times 0.3010$$

$$k = 0.6932\,min^{-1}$$

$$k = \left(\frac{0.6932}{60}\right) s^{-1}$$

$$k = 1.153 \times 10^{-2}\ s^{-1}$$

19. தீர்வு

$$k = A\ e^{-\left(\frac{E_a}{RT}\right)}$$

$$k = 1.6 \times 10^{13} s^{-1} e^{-\left(\frac{200\times103\ J\ mol^{-1}}{8.314 JK^{-1}mol^{-1}\times600 K}\right)}$$

$$k = 1.6 \times 10^{13} s^{-1}\ e^{-(40.1)}$$

$$k = 1.6 \times 10^{13} s^{-1} \times 3.8 \times 10^{-18}$$

$$k = 6.21 \times 10^{-5} s^{-1}$$

20. தீர்வு

வினைவேகம் $= k[x]^n [y]^m$

$$0.15 = k[0.2]^n [0.02]^m \text{ ———(1)}$$

$$0.30 = k[0.4]^n [0.02]^m \text{ ———(2)}$$

$$1.20 = k[0.4]^n [0.08]^m \text{ ———(3)}$$

$$(3) \Big/ (2)$$

$$\frac{1.2}{0.3} = \frac{k[0.4]^n [0.08]^m}{k[0.4]^n [0.02]^m}$$

$$4 = \left(\frac{[0.08]}{[0.02]}\right)^m$$

$$4 = (4)^m$$

$$\therefore\ m = 1$$

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